

WEEK 5

S.SANDHYA SHINDE

Healthcare Dataset

The healthcare_dataset.csv file was imported into Power BI Desktop.

healthcare_dataset.csv

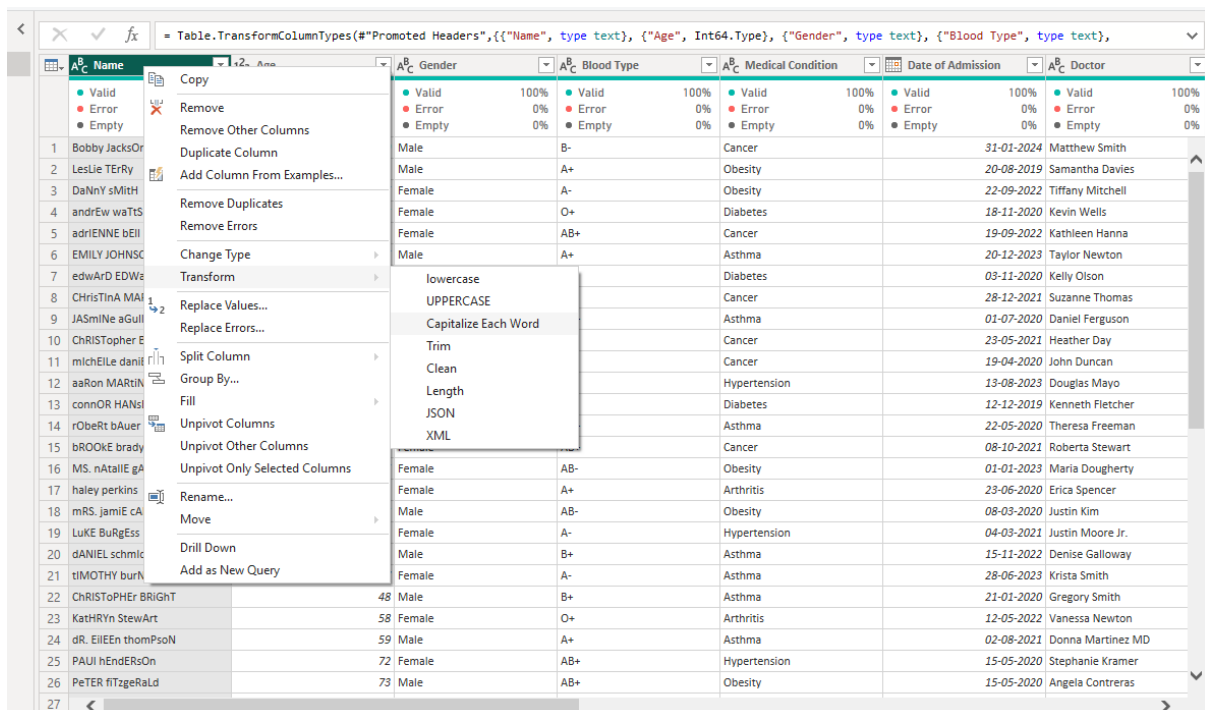
File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance
Bobby JacksOn	30	Male	B-	Cancer	31-01-2024	Matthew Smith	Sons and Miller	Blue Cross
LesLie TErRy	62	Male	A+	Obesity	20-08-2019	Samantha Davies	Kim Inc	Mechanical
DaNnY sMitH	76	Female	A-	Obesity	22-09-2022	Tiffany Mitchell	Cook PLC	Aetna
andrEw waTtS	28	Female	O+	Diabetes	18-11-2020	Kevin Wells	Hernandez Rogers and Vang,	Mechanical
adriENNE bEll	43	Female	AB+	Cancer	19-09-2022	Kathleen Hanna	White-White	Aetna
EMILY JOHNSOn	36	Male	A+	Asthma	20-12-2023	Taylor Newton	Nunez-Humphrey	UnitedHealth
edwArD EDWaRDs	21	Female	AB-	Diabetes	03-11-2020	Kelly Olson	Group Middleton	Mechanical
CHrisTinA MARTinez	20	Female	A+	Cancer	28-12-2021	Suzanne Thomas	Powell Robinson and Valdez,	Cigna
JASmiNe aGullaR	82	Male	AB+	Asthma	01-07-2020	Daniel Ferguson	Sons Rich and	Cigna
ChRISTopher BerG	58	Female	AB-	Cancer	23-05-2021	Heather Day	Padilla-Walker	UnitedHealth
mlchEiLe daniELs	72	Male	O+	Cancer	19-04-2020	John Duncan	Schaefer-Porter	Mechanical
aaRon MARTineZ	38	Female	A-	Hypertension	13-08-2023	Douglas Mayo	Lyons-Blair	Mechanical
connOR HANsEn	75	Female	A+	Diabetes	12-12-2019	Kenneth Fletcher	Powers Miller, and Flores	Cigna
rObert bAuer	68	Female	AB+	Asthma	22-05-2020	Theresa Freeman	Rivera-Gutierrez	UnitedHealth
bROOkE brady	44	Female	AB+	Cancer	08-10-2021	Roberta Stewart	Morris-Arellano	UnitedHealth
MS. nAtalie gAMble	46	Female	AB-	Obesity	01-01-2023	Maria Dougherty	Cline-Williams	Blue Cross
haley perkins	63	Female	A+	Arthritis	23-06-2020	Erica Spencer	Cervantes-Wells	UnitedHealth
mRS. jamiE cAMPBELl	38	Male	AB-	Obesity	08-03-2020	Justin Kim	Torres, and Harrison Jones	Cigna
LuKE BuRgEss	34	Female	A-	Hypertension	04-03-2021	Justin Moore Jr.	Houston PLC	Blue Cross
dANIEL schmidt	63	Male	B+	Asthma	15-11-2022	Denise Galloway	Hammond Ltd	Cigna

Extract Table Using Examples | Load | Transform Data | Cancel

2. Data Cleaning and Data Type Verification

The dataset was opened in Power Query Editor. The columns were checked and the required data types were verified. The data quality was also checked for valid, error, and empty values.



	A _C Name	A _C Gender	A _C Blood Type	A _C Medical Condition	A _C Date of Admission	A _C Doctor
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	Bobby JacksOn	Male	B-	Cancer	31-01-2024	Matthew Smith
2	Leslie TerRy	Male	A+	Obesity	20-08-2019	Samantha Davies
3	DaNnY sMitH	Female	A-	Obesity	22-09-2022	Tiffany Mitchell
4	andREW waTts	Female	O+	Diabetes	18-11-2020	Kevin Wells
5	adriENNE bEll	Female	AB+	Cancer	19-09-2022	Kathleen Hanna
6	EMILY JOHNSC	Male	A+	Asthma	20-12-2023	Taylor Newton
7	edwArD EDWa			Diabetes	03-11-2020	Kelly Olson
8	ChrisTinA MAI			Cancer	28-12-2021	Suzanne Thomas
9	JASmiNe aGull			Asthma	01-07-2020	Daniel Ferguson
10	ChRISTopher E			Cancer	23-05-2021	Heather Day
11	michElLe daniE			Cancer	19-04-2020	John Duncan
12	aaRon MARtin			Hypertension	13-08-2023	Douglas Mayo
13	connOR HANsI			Diabetes	12-12-2019	Kenneth Fletcher
14	rObert bAuer			Asthma	22-05-2020	Theresa Freeman
15	bROOkE brady			Cancer	08-10-2021	Roberta Stewart
16	MS. nAtalie gA	Female	AB-	Obesity	01-01-2023	Maria Dougherty
17	haley perkins	Female	A+	Arthritis	23-06-2020	Erica Spencer
18	mRS. jamiE cA	Male	AB-	Obesity	08-03-2020	Justin Kim
19	LuKE BuRgEss	Female	A-	Hypertension	04-03-2021	Justin Moore Jr.
20	dANIEL schmId	Male	B+	Asthma	15-11-2022	Denise Galloway
21	tIMOTHY burN	Female	A-	Asthma	28-06-2023	Krista Smith
22	ChRISToPHER BRIGHt	48 Male	B+	Asthma	21-01-2020	Gregory Smith
23	KathRyn StewArT	58 Female	O+	Arthritis	12-05-2022	Vanessa Newton
24	dR. ElIEEn thomPsoN	59 Male	A+	Asthma	02-08-2021	Donna Martinez MD
25	PAUI hEndERsOn	72 Female	AB+	Hypertension	15-05-2020	Stephanie Kramer
26	PeTER fITZgeRaLd	73 Male	AB+	Obesity	15-05-2020	Angela Contreras
27						

3. Creating Age Group

An Age Group column was created using a Conditional Column.

The following age groups were created:

- 0-18
- 19-35
- 36-60
- 60 above

Add Conditional Column



Add a conditional column that is computed from the other columns or values.

New column name

Age Group

	Column Name	Operator	Value ①		Output ①	
If	Age	is less than or equ...	ABC 123 18	Then	ABC 123 0-18	...
Else If	Age	is less than or equ...	ABC 123 35	Then	ABC 123 19-35	
Else If	Age	is less than or equ...	ABC 123 60	Then	ABC 123 36-60	

Add Clause

Else ①

ABC 123

60 above

OK

Cancel

4. Creating Dim_Patient

A reference query was created to prepare the Dim_Patient table.

The table contains:

- PatientID
- Name
- Age
- Age Group
- Gender
- Blood Type

Duplicate patient records were removed, and PatientID was created using an Index column.

Creating Dim_Patient

Queries [5]

healthcare_dataset

Dim_Patient

Dim_Admission

Fact_Billing

Measures Table

Table.Distinct(#'Reordered Columns', {'Name', 'Age', 'Age Group', 'Gender', 'Blood Type'})

PatientID	Name	Age	Age Group	Gender	Blood Type
1	Bobby Jackson	30	19-35	Male	B-
2	Leslie Terry	62	60 above	Male	A+
3	Danny Smith	76	60 above	Female	A-
4	Andrew Watts	28	19-35	Female	O+
5	Adrienne Bell	43	36-60	Female	AB+
6	Emily Johnson	36	36-60	Male	A+
7	Edward Edwards	21	19-35	Female	AB-
8	Christina Martinez	20	19-35	Female	A+
9	Jasmine Aguilar	82	60 above	Male	AB+
10	Christopher Berg	58	36-60	Female	AB-
11	Michelle Daniels	72	60 above	Male	O+
12	Aaron Martinez	38	36-60	Female	A-
13	Connor Hansen	75	60 above	Female	A+
14	Robert Bauer	68	60 above	Female	AB+
15	Brooke Brady	44	36-60	Female	AB+
16	Ms. Natalie Gamble	46	36-60	Female	AB-
17	Haley Perkins	63	60 above	Female	A+
18	Mrs. Jamie Campbell	38	36-60	Male	AB-
19	Luke Burgess	34	19-35	Female	A-
20	Daniel Schmidt	63	60 above	Male	B+
21	Timothy Burns	67	60 above	Female	A-
22	Christopher Bright	48	36-60	Male	B+
23	Kathryn Stewart	58	36-60	Female	O+
24	Dr. Eileen Thompson	59	36-60	Male	A+
25	Paul Henderson	72	60 above	Female	AB+
26	Peter Fitzgerald	73	60 above	Male	AB+
27	Cathy Small	51	36-60	Female	O-
28	Mr. Kenneth Moore	34	19-35	Female	A+
29	Mary Hunter	38	36-60	Female	O-
30	Joshua Oliver	63	60 above	Female	B+
31	Thomas Martinez	34	19-35	Male	B-

Query Settings

Properties

Name

Dim_Patient

All Properties

Applied Steps

Source

Removed Other Columns

Removed Columns

Added Index

Renamed Columns

Reordered Columns

Removed Duplicates

5. Creating Dim_Admission

A reference query was created to prepare the Dim_Admission table.

The table contains:

- AdmissionID
- Medical Condition
- Admission Type
- Doctor
- Hospital
- Room Number
- Medication
- Test Results

Duplicate admission records were removed, and AdmissionID was created using an Index column.

Creating Dim_Admission

Table: ReorderColumns(*Renamed Columns*, ("AdmissionID", "Medical Condition", "Admission Type", "Doctor", "Hospital", "Room Number"))

AdmissionID	Medical Condition	Admission Type	Doctor	Hospital	Room Number
1	Cancer	Urgent	Matthew Smith	Sons and Miller	
2	Obesity	Emergency	Samantha Davies	Kim Inc	
3	Obesity	Emergency	Tiffany Mitchell	Cook PLC	
4	Diabetes	Elective	Kevin Wells	Hernandez Rogers and Vang,	
5	Cancer	Urgent	Kathleen Hanna	White-White	
6	Asthma	Urgent	Taylor Newton	Nunez-Humphrey	
7	Diabetes	Emergency	Kelly Olson	Group Middleton	
8	Cancer	Emergency	Suzanne Thomas	Powell Robinson and Valdez,	
9	Asthma	Elective	Daniel Ferguson	Sons Rich and	
10	Cancer	Elective	Heather Day	Padilla-Walker	
11	Cancer	Urgent	John Duncan	Schaefer-Porter	
12	Hypertension	Urgent	Douglas Mayo	Lyons-Blair	
13	Diabetes	Emergency	Kenneth Fletcher	Powers Miller, and Flores	
14	Asthma	Urgent	Theresa Freeman	Rivera-Gutierrez	
15	Cancer	Urgent	Roberta Stewart	Morris-Arellano	
16	Obesity	Elective	Maria Dougherty	Cline-Williams	
17	Arthritis	Elective	Erica Spencer	Cervantes-Wells	
18	Obesity	Urgent	Justin Kim	Torres, and Harrison Jones	
19	Hypertension	Elective	Justin Moore Jr.	Houston PLC	
20	Asthma	Elective	Denise Galloway	Hammond Ltd	
21	Asthma	Elective	Krista Smith	Jones LLC	
22	Asthma	Urgent	Gregory Smith	Williams-Davis	
23	Arthritis	Urgent	Vanessa Newton	Clark-Mayo	
24	Asthma	Urgent	Donna Martinez MD	and Sons Smith	
25	Hypertension	Emergency	Stephanie Kramer	Wilson Group	
26	Obesity	Urgent	Angela Contreras	Garner-Bowman	
27	Asthma	Elective	Wendy Glenn	Brown, and Jones Weaver	
28	Diabetes	Emergency	James Ellis	Serrano-Dixon	
29	Hypertension	Emergency	Jared Bruce Jr.	Gardner-Miller	
30	Hypertension	Elective	Brandy Mitchell	Guerrero-Boone	
31	Asthma	Urgent	Isaac Hough	Hart Ltd	

6. Creating Fact_Billing

The Fact_Billing table was prepared with the required billing and admission information.

The final columns are:

- PatientKey
- AdmissionKey
- Date of Admission
- Discharge Date
- Insurance Provider
- Billing Amount

Creating Fact_Billing

The screenshot displays a data modeling tool interface. On the left, a 'Queries' pane lists 'Fact_Billing' as the selected query. The main area shows a table with columns: PatientKey, AdmissionKey, Date of Admission, Discharge Date, Insurance Provider, and Billing Amount. Each column has a data type and a 'Valid' status indicator. The 'Fact_Billing' table is linked to 'Dim_Patient' and 'Dim_Admission'. The 'Query Settings' pane on the right shows the 'Fact_Billing' query with a 'Name' field and a 'Reordered Columns' section.

Table	Column	Data Type	Valid
Fact_Billing	PatientKey	Integer	Valid
Fact_Billing	AdmissionKey	Integer	Valid
Fact_Billing	Date of Admission	Date	Valid
Fact_Billing	Discharge Date	Date	Valid
Fact_Billing	Insurance Provider	Text	Valid
Fact_Billing	Billing Amount	Decimal	Valid

7. Merging Fact_Billing with Dim_Admission

The Fact_Billing table was merged with Dim_Admission using the matching fields:

- Medical Condition
- Admission Type
- Doctor
- Hospital
- Room Number
- Medication
- Test Results

The AdmissionID was expanded and renamed as AdmissionKey.

Merge Fact_Billing with Dim_Admission

Merge

Select a table and matching columns to create a merged table.

Fact_Billing



	Room Number	Admission Type	Discharge Date	Medication	Test Results	Age Group	Patient Key
1	328	Urgent	02-02-2024	Paracetamol	Normal	19-35	1
9	265	Emergency	26-08-2019	Ibuprofen	Inconclusive	60 above	2
8	205	Emergency	07-10-2022	Aspirin	Normal	60 above	3
1	450	Elective	18-12-2020	Ibuprofen	Abnormal	19-35	4

Dim_Admission



	AdmissionID	Medical Condition	Admission Type	Doctor	Hospital	Room Number
	1	Cancer	Urgent	Matthew Smith	Sons and Miller	328
	2	Obesity	Emergency	Samantha Davies	Kim Inc	265
	3	Obesity	Emergency	Tiffany Mitchell	Cook PLC	205
	4	Diabetes	Elective	Kevin Wells	Hernandez Rogers and Vang,	450

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

▸ Fuzzy matching options

✓ The selection matches 55500 of 55500 rows from the first table.

OK

Cancel

8. Merging Fact_Billing with Dim_Patient

The Fact_Billing table was merged with Dim_Patient using:

- Name
- Age
- Gender
- Blood Type

The PatientID was expanded and renamed as PatientKey.

Merge Fact_Billing with Dim_Patient

Merge

Select a table and matching columns to create a merged table.

Fact_Billing

Name	1	Age	2	Gender	3	Blood Type	4	Medical Condition	Date of Admission	Doctor	
Bobby Jackson		30		Male		B-		Cancer	31-01-2024	Matthew Smith	Sons and
Leslie Terry		62		Male		A+		Obesity	20-08-2019	Samantha Davies	Kim Inc
Danny Smith		76		Female		A-		Obesity	22-09-2022	Tiffany Mitchell	Cook PLC
Andrew Watts		28		Female		O+		Diabetes	18-11-2020	Kevin Wells	Hernand

Dim_Patient

PatientID	Name	1	Age	2	Age Group	Gender	3	Blood Type	4
1	Bobby Jackson		30		19-35	Male		B-	
2	Leslie Terry		62		60 above	Male		A+	
3	Danny Smith		76		60 above	Female		A-	
4	Andrew Watts		28		19-35	Female		O+	
5	Adrienne Bell		43		36-60	Female		AB+	

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

▸ Fuzzy matching options

✓ The selection matches 55500 of 55500 rows from the first table.

OK

Cancel

9. Creating Data Model Relationships

The required relationships were created between the dimension tables and the Fact_Billing table.

Relationship 1:

Dim_Patient[PatientID] → Fact_Billing[PatientKey]

- Cardinality: One-to-Many (1:*)
- Cross-filter direction: Single
- Active: Yes

Relationship 2:

Dim_Admission[AdmissionID] → Fact_Billing[AdmissionKey]

- Cardinality: One-to-Many (1:*)
- Cross-filter direction: Single
- Active: Yes

10. Final Data Model

